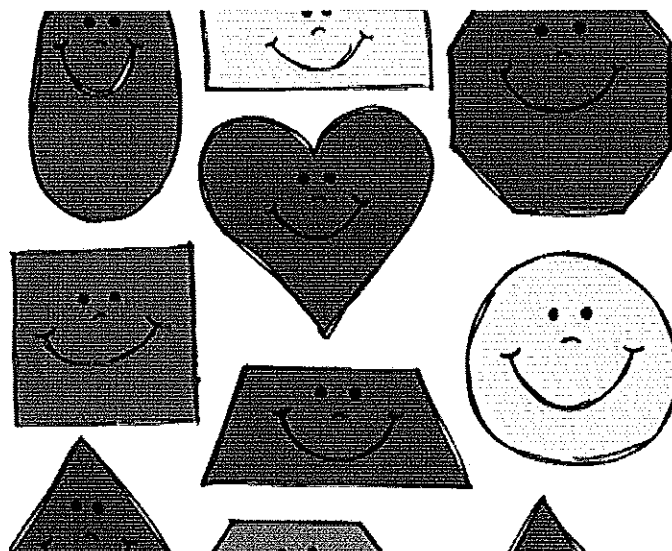


**MATHEMATICS
YEAR 8**

Problem 3 – Angles in a Polygon

Semester 2 2014



Information for Students

Working time:	40 minutes
Marks available:	18 marks

Interior Angles In Polygons

You are going to investigate angles inside polygons and look at what the sum of the interior angles are when you add them up. We call the angles inside a polygon 'Interior Angles'.

Follow the instructions below to fill in the table and then answer the questions at the bottom of the page.

INSTRUCTIONS

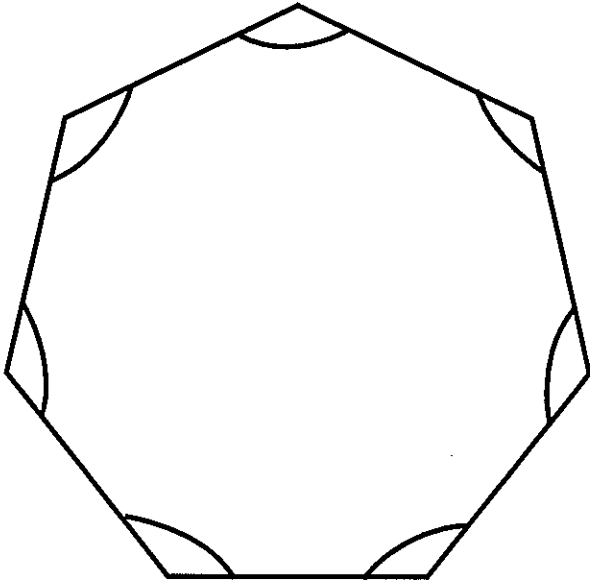
1. Measure all the interior angles inside the shapes on your worksheets using a protractor.
2. Add all the interior angles up for each shape and write the answer under each shape.
3. Work out the minimum number of triangles to check your answers for the sum of the interior angles using the 'Minimum number of Triangles' worksheet provided.

Round all your answers in the 'sum of interior angles' column to the nearest 10 degrees.

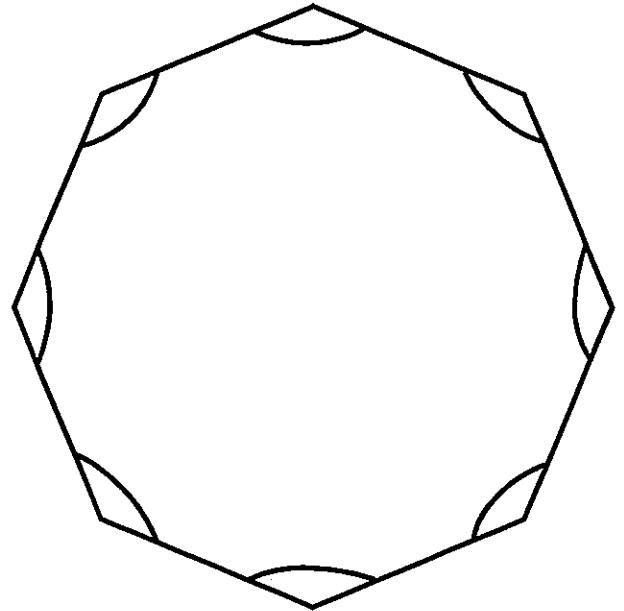
Shape name	Number of Sides	Min number of triangles	Sum of Interior Angles
Triangle	3	1	180°
Quadrilateral	4	2	360°
Pentagon	5	3	540°
Hexagon	6	4	720°
heptagon	7	5	900°
Octagon	8	6	1080°
Nonagon	9	7	1260°
Decagon	10	8	1440°

(10 marks)

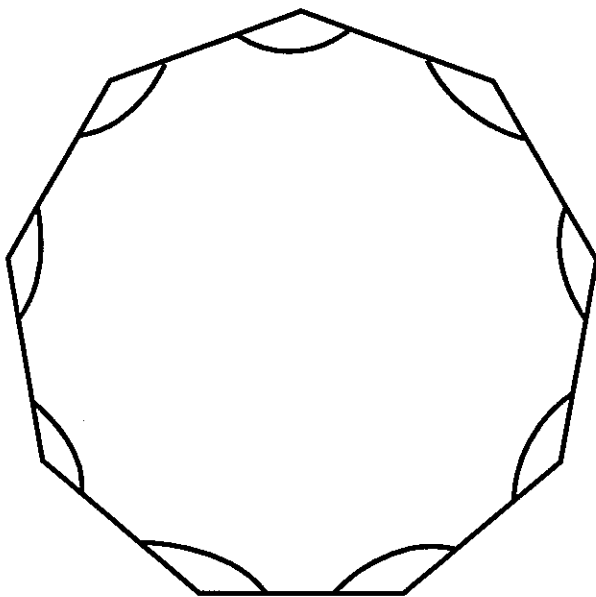
$\frac{1}{2}$ marks each



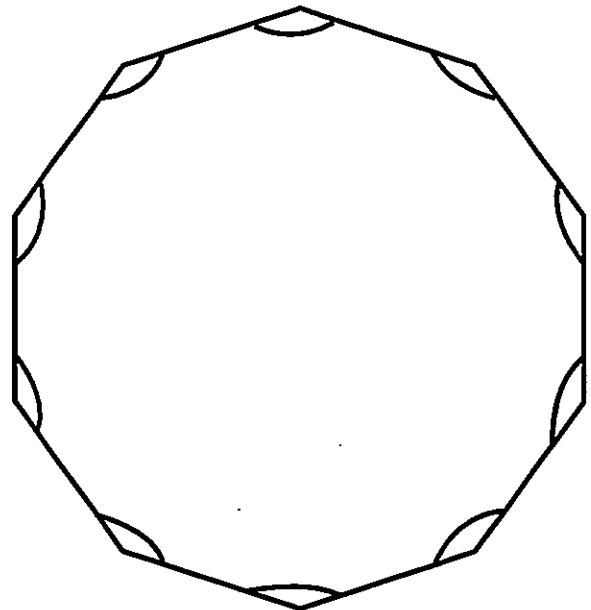
Sum of interior angles =500.....



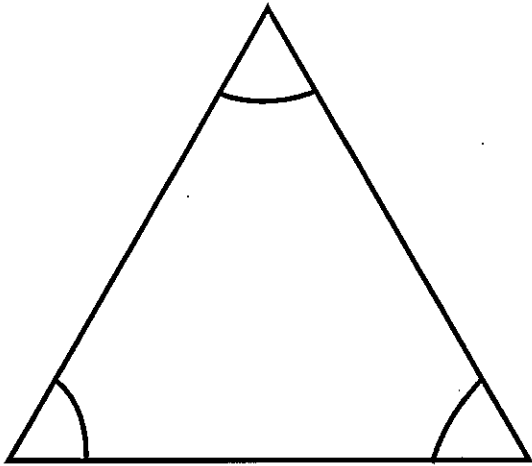
Sum of interior angles =1080.....



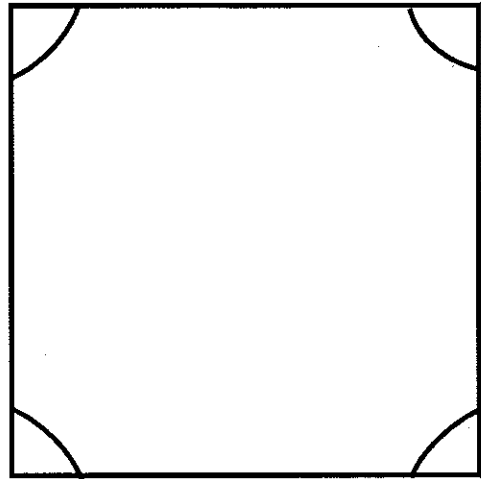
Sum of interior angles =1260.....



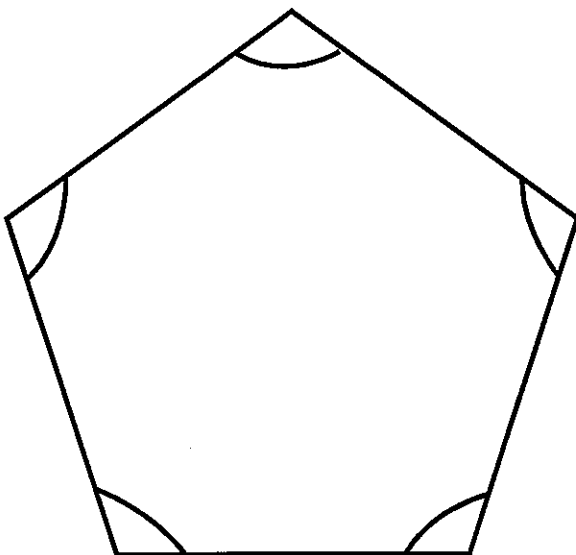
Sum of interior angles =1440.....



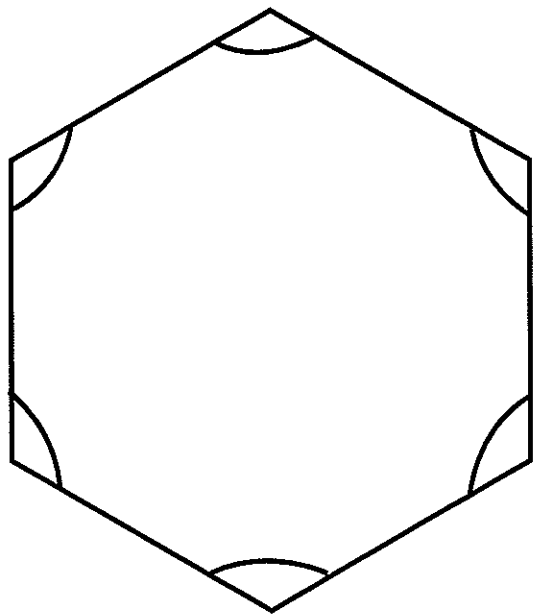
Sum of interior angles = 180°



Sum of interior angles = 360°



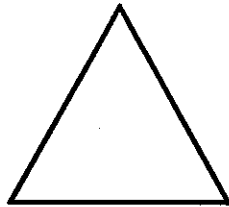
Sum of interior angles = 540°



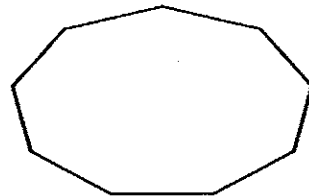
Sum of interior angles = 720°

Minimum number of triangles in a shape

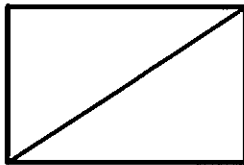
Instructions – Rule a line from one vertex (corner) to all the other vertexes to figure out how many triangles are in each shape. Remember that each triangle has angles that add up to 180° . The first two have been done for you.



$$= 180^\circ$$

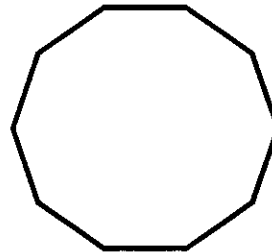


$$7 \times 180^\circ = 1260^\circ$$

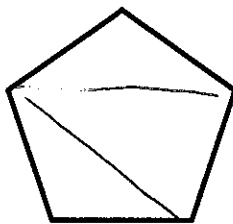


$$= 2 \text{ triangles} = 2 \times 180^\circ$$

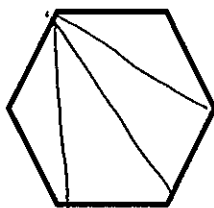
$$= 360^\circ$$



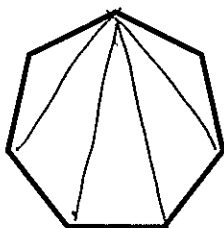
$$8 \times 180^\circ = 1440^\circ$$



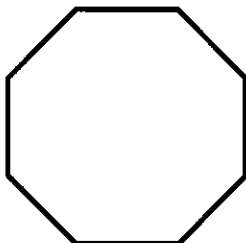
$$3 \times 180^\circ = 540^\circ$$



$$4 \times 180^\circ = 720^\circ$$



$$5 \times 180^\circ = 900^\circ$$



$$6 \times 180^\circ = 1080^\circ$$

Questions

1. Can you spot a pattern in how the sum of the interior angles increases as another side is added to the shape? (1 mark)

goes up by 180°

2. Can you find a rule to explain how to get from the number of sides to the sum of the interior angles? (2 mark)

$$(\text{sides} - 2) \times 180^\circ$$

① ①

3. Can you use your rule to work out what the sum of the interior angles would be for a 12 sided shape? (2 mark)

$$(12 - 2) \times 180 = 1800^\circ$$

① ②

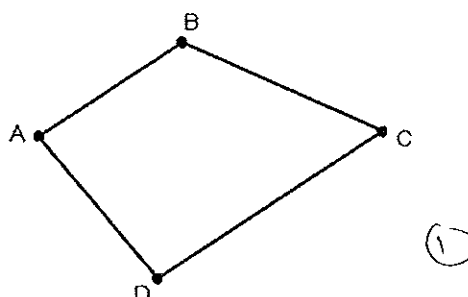
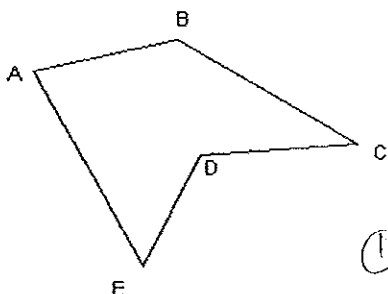
① for answer but no calculation

4. All the shapes in your investigation were regular (all sides and angles the same length).

Would the answers be the same for irregular polygons?

(2 marks)

Work out the angles of these two irregular shapes to check if the answers would be the same.



Sum of angles for 5 sided polygon? 540°

Sum of angles for a Quadrilateral? 360°

Would the answers be the same? yes

5. For a polygon with n sides, write an expression to calculate the sum of the interior angles?

(1 mark)

$$(n-2) \times 180$$

(1)

End of Problem